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Hiring Manager

The University of Alabama

Dear Hiring Manager,

My name is Ajay Venkatesh. I'm finishing my M.S. in Computer Engineering at NYU, with production software experience at PwC in Bengaluru, a summer SDE internship at Amazon, and ongoing on-campus work at NYU's Novel AI Technologies. I'm writing about the Data Scientist I role supporting water security research at the University of Alabama.

The role's emphasis on **data management and integration**, plus building **ETL** processes into shared data repositories for university and federal research partners, lines up closely with my engineering work. At NYU's Novel AI Technologies I engineered **ETL pipelines** transforming **NoSQL** into **PostgreSQL** with concurrency control and locking mechanisms, processing hundreds of thousands of records, and powered **AWS QuickSight** dashboards for real-time analytics. At PwC I owned a distributed backend on **Microsoft Azure** over **SQL Server**, writing complex queries, stored procedures, and parallelized batch ingestion that materially cut execution time on operational feeds.

On database administration and statistical analysis: my graduate work spans **Big Data**, **Machine Learning**, and **Deep Learning** coursework, plus a **PySpark / HDFS** project over millions of flight records with EDA, multivariate analysis, and feature engineering. I'm comfortable across **MySQL**, **PostgreSQL**, **MongoDB**, and **DynamoDB**, and at home with the data-quality work that scientific repositories demand.

On cross-functional research collaboration: I've owned features end-to-end alongside business stakeholders at PwC and partnered with researchers at NYU's Novel AI Technologies on an NSF-funded health platform. Translating research needs into reproducible data infrastructure is the kind of work where my engineering background compounds.

Stack-wise: **Python**, **MySQL**, **PostgreSQL**, **ETL** workflows, **Pandas**, **NumPy**, **PySpark**, **Apache Spark**, plus visualization through **Power BI** and **QuickSight**. Comfortable in Linux and at home with statistical methods.

What pulls me toward this role is the mission. Water security research at the federal-state-university intersection runs on careful, reproducible data infrastructure that compounds across studies, and contributing to research outcomes that matter beyond any single product is meaningful work.

I'd welcome the chance to discuss further. Thanks for considering my application.

Sincerely,

Ajay Venkatesh